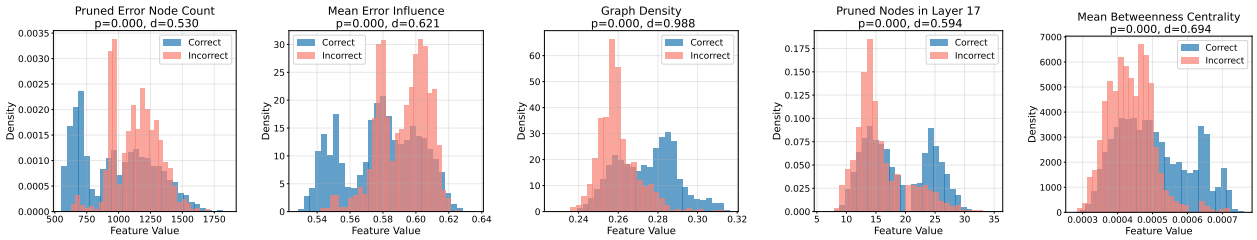


Figure 6 Topological Fingerprints of Error on Arithmetic. Distributions of five selected graph features for correct (blue) vs. incorrect (red) reasoning steps. The visual separation is statistically significant for each feature shown (independent t-test, $p < 0.001$) and represents a medium-to-large effect size (Cohen's d). This provides quantitative evidence that attribution graphs contain a clear, separable structural signal of a computation's integrity.

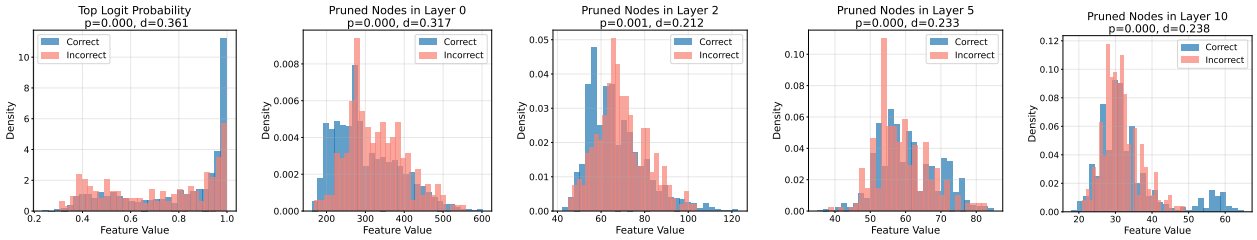
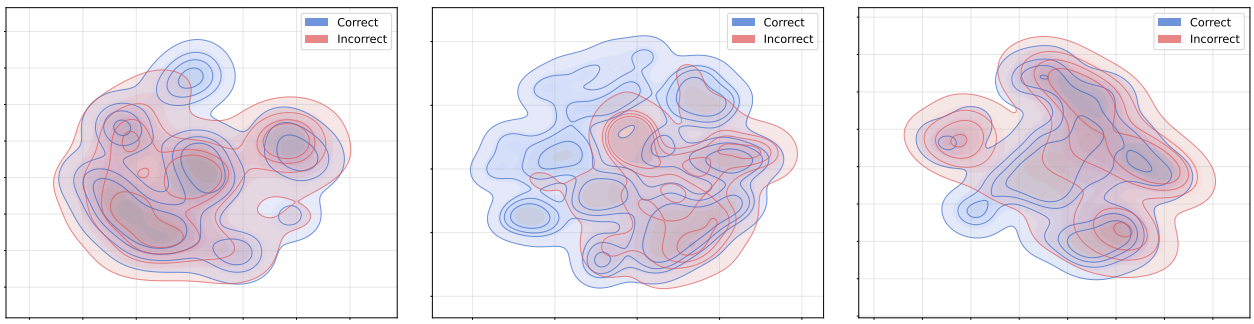


Figure 7 Topological Fingerprints of Error on Boolean. Distributions of five selected graph features for correct (blue) vs. incorrect (red) reasoning steps. The visual separation is statistically significant for each feature shown (independent t-test, $p < 0.001$) and represents a medium-to-large effect size (Cohen's d). This provides quantitative evidence that attribution graphs contain a clear, separable structural signal of a computation's integrity.

We demonstrate another casual intervention with a concrete case study on the arithmetic task, where we correct a subtle mathematical parsing error not by suppressing a faulty feature, but by *amplifying* a correct one. We present the model with the expression `(-(5+(4*9)))`. As shown in Table 19, the model initially misinterprets the leading unary minus, treating it as a subtraction from an implicit -1 and producing the incorrect result of -42. Our CRV classifier flagged this step as incorrect, and its feature importance analysis suggested that the error stemmed from the absence of a strong "multiplication" signal.

Hypothesizing that the model failed to recognize the unary minus as a multiplication by -1, we performed a direct intervention. We identified a transcoder feature from the graph whose semantics corresponded to multiplication and, using a forward hook, manually amplified its activation value just before the model generated its erroneous step. The effect was immediate and successful. As shown in the "After Intervention" column, this boost was sufficient to steer the model onto the correct computational path. It now correctly multiplies by -1 and arrives at the final answer of -41.



(a) Boolean **(b)** Arithmetic **(c)** GSM8K

Figure 8 Distributions of features after t-SNE for correct (blue) vs. incorrect (red) reasoning steps. The separation validates that CRV captures a distinct structural fingerprint of error.

Before Intervention (Incorrect)	**After Intervention (Correct)**
Evaluate the arithmetic expression below. (- (5 + (4 * 9))) To evaluate this expression, we need to follow the order of operations (PEMDAS): 1. Multiply 4 and 9: 4 * 9 = 36 2. Add 5 and 36: 5 + 36 = 41 3. Subtract 41 from -1: -1 - 41 = -42 The final answer is -42.	Evaluate the arithmetic expression below. (- (5 + (4 * 9))) To evaluate this expression, we need to follow the order of operations (PEMDAS): 1. Multiply 4 and 9: 4 * 9 = 36 2. Add 5 and 36: 5 + 36 = 41 3. Multiply -1 by 41: -1 * 41 = -41 The final answer is: -41.

Table 19 A successful causal intervention via feature amplification. A subtraction error is corrected by amplifying a multiplication feature, steering the model to the correct reasoning path.

D Usage of Language Language Models

In the preparation of this manuscript, we utilized LLMs as writing assistants. Their role was strictly limited to improving the clarity, conciseness, and grammatical correctness of the text. The authors take full responsibility for all content and any remaining errors.

E Limitations

Our work introduces a new methodology for the scientific analysis of reasoning, and its limitations are intrinsically linked to its design as a white-box, mechanistic instrument.

Computational Intensity. A primary limitation of CRV is its computational intensity. The process, which involves training a suite of transcoders, replacing model modules, and constructing a detailed attribution graph for every reasoning step, is orders of magnitude more resource-intensive than black-box or gray-box verification methods. This cost is a direct consequence of our white-box approach, which prioritizes mechanistic depth over practical efficiency. As such, CRV in its current form is positioned as a scientific tool for deep analysis, not as a scalable, real-time verifier for production systems.

Aggregative vs. Feature-Level Analysis. The feature set used by CRV is primarily aggregative; it captures statistical and topological properties of the graph, such as node counts, influence scores, and density. As an early work, it does not yet fully exploit the semantic content of the individual transcoder features that constitute the graph's nodes. For instance, our current classifier learns statistical correlations over the entire feature set; it does not reason symbolically about whether a specific feature for numerical addition is appropriately activated by numerical inputs. This represents a significant opportunity. A promising future direction lies in developing more sophisticated classifiers or rule-based systems that operate directly on the semantics of these disentangled features, paving the way for a new class of neuro-symbolic verifiers.

Generalizability of Error Signatures. Our empirical results are based on a single model family (Llama 3.1) at the 8B scale. Whether the precise structural fingerprints we identified generalize to different architectural paradigms, such as Mixture-of-Experts, or across significant model scales (e.g., 70B and larger) remains an open question. Furthermore, as our cross-domain experiments revealed, the error signatures are highly domain-specific. Our work provides a strong foundation and a methodology for discovering these signatures, but further studies are needed to determine if more universal principles of computational failure exist.

Fidelity of Interpretability Tools. The validity of our analysis is contingent on the quality and fidelity of the underlying interpretability tools. The features identified by our transcoders, while demonstrably useful, represent one possible sparse basis and are not exhaustive. Similarly, the attribution method provides a powerful but ultimately incomplete approximation of the true information flow within the model. Future improvements in these foundational techniques, such as the development of more faithful sparse autoencoders or more precise attribution methods, will directly enhance the resolution and reliability of analyses like ours.